See appended validity comments for suggested wording changes.

Bounding the Infinite-Bagged RSM Remainder under Common Loading

A randomized-linear-algebra/coherence argument

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1 Purpose

This note gives a clean way to bound the infinite-bagged RSM remainder

$$R_k^{\mathrm{bag},\infty} = Q_k^{\mathrm{bag},\infty} - Q^{\mathrm{opt}},$$

under the common-loading heteroskedastic forecast-error structure. The goal is not to assume a functional form such as C/k^2, but to show when such a rate is theoretically justified. The randomized-linear-algebra (RandNLA) idea enters through leverage coherence: uniform random subspace sampling is close to leverage sampling only when the leverage/oracle weights are not too concentrated. When leverage is highly concentrated, the safe bound is the Jensen bound

$$0 \le R_k^{\mathrm{bag},\infty} \le Q_k^{\mathrm{sub}} - Q^{\mathrm{opt}}.$$

2 Common-loading setup and exact remainder

Assume the common-loading covariance structure

$$\Sigma_e = q\iota\iota' + D, \qquad D = \mathrm{diag}(\sigma_1^2, \dots, \sigma_m^2), \qquad \tau_i = \sigma_i^{-2} > 0.$$

Let

$$A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}, \qquad \ell_i = \frac{\tau_i}{A_m}, \qquad \sum_{i=1}^m \ell_i = 1.$$

Under common loading, the full oracle combination weight is

$$w_i^* = \frac{\tau_i}{A_m} = \ell_i,$$

so the oracle weights are exactly the normalized precision scores. This is the link to leverage-score sampling: in this model, the leverage score of coordinate i is ℓ_i.

For a subset S of size k, define its sampled leverage mass

$$L_S = \sum_{j \in S} \ell_j = \frac{A_S}{A_m}, \qquad A_S = \sum_{j \in S} \tau_j.$$

The embedded subset weight is

$$v_{S,i} = \mathbf{1}\{i \in S\}\frac{\tau_i}{A_S} = \mathbf{1}\{i \in S\}\frac{\ell_i}{L_S}.$$

Therefore the infinite-bagged weight is

$$\bar{v}_{k,i} = \mathbb{E}_S \left[\mathbf{1}\{i \in S\} \frac{\ell_i}{L_S} \right].$$

Since $\bar{v}_k' \iota = 1$, the exact excess-risk identity is

$$R_k^{\mathrm{bag},\infty} = (\bar{v}_k - w^*)' \Sigma_e (\bar{v}_k - w^*) = \sum_{i=1}^m \frac{(\bar{v}_{k,i} - w_i^*)^2}{\tau_i}.$$

Because $w_i^* = \ell_i$ and $\tau_i = A_m \ell_i$, this can be rewritten as

$$\boxed{R_k^{\mathrm{bag},\infty} = \frac{1}{A_m} \sum_{i=1}^m \ell_i \left(\frac{\bar{v}_{k,i}}{\ell_i} - 1 \right)^2.} \tag{1}$$

This is the key RandNLA form: R_k is a weighted squared error from estimating the leverage/oracle vector by uniform subset sampling.

3 The coherence parameter

Define the leverage coherence

$$\kappa = m \max_i \ell_i = \frac{\tau_{\max}}{\bar{\tau}}.$$

Low coherence means that no coordinate dominates the oracle weight. High coherence means that one or a few forecasts carry a large fraction of the oracle weight.

It is also useful to define the normalized precision dispersion

$$h_i = m\ell_i - 1 = \frac{\tau_i - \bar{\tau}}{\bar{\tau}}, \qquad V_\delta = \frac{1}{m} \sum_{i=1}^m h_i^2 = m \sum_{i=1}^m \left(\ell_i - \frac{1}{m} \right)^2.$$

The term V_δ is the dispersion of leverage scores around the uniform design.

4 Low-coherence/balanced-leverage bound

A purely upper-coherence condition controls the largest leverage score. To control reciprocal sampled masses uniformly, we use a balanced-leverage condition.

Assumption 1 (Balanced leverage)**.** *There exist constants $0 < c \le C < \infty$ such that, for all m and all i,*

$$\frac{c}{m} \le \ell_i \le \frac{C}{m}.$$

Equivalently,

$$c \le \frac{\tau_i}{\bar{\tau}} \le C.$$

This is a standard low-coherence/no-dominant-coordinate condition strengthened with a lower bound to prevent sampled leverage mass from becoming arbitrarily small.

Let $p = k/m$. Conditional on $i \in S$, write

$$L_{i,R} = \ell_i + \sum_{j \in R} \ell_j,$$

where R is a subset of size $k-1$ drawn uniformly without replacement from $\{1,\dots,m\}\setminus\{i\}$. Then

$$\frac{\bar{v}_{k,i}}{\ell_i} = p\,\mathbb{E}_R\left[\frac{1}{L_{i,R}}\right].$$

The conditional mean is

$$\mu_i = \mathbb{E}_R[L_{i,R}] = \ell_i + (k-1)\frac{1-\ell_i}{m-1} = p + \frac{m-k}{m-1}\left(\ell_i - \frac{1}{m}\right).$$

Also define

$$z_k = \frac{1}{k} - \frac{1}{m} = \frac{m-k}{km}.$$

Lemma 1 (Reciprocal expansion with explicit remainder)**.** *For each i,*

$$\mathbb{E}_R\left[\frac{1}{L_{i,R}}\right] = \frac{1}{\mu_i} + \mathbb{E}_R\left[\frac{(L_{i,R}-\mu_i)^2}{L_{i,R}\mu_i^2}\right].$$

Consequently,

$$\frac{\bar{v}_{k,i}}{\ell_i} - 1 = \left(\frac{p}{\mu_i} - 1\right) + p\,\mathbb{E}_R\left[\frac{(L_{i,R}-\mu_i)^2}{L_{i,R}\mu_i^2}\right].$$

Proof. Use the exact identity

$$\frac{1}{x} = \frac{1}{\mu} - \frac{x-\mu}{\mu^2} + \frac{(x-\mu)^2}{x\mu^2}.$$

Take expectations with $x = L_{i,R}$ and $\mu = \mu_i$. Since $\mathbb{E}_R[L_{i,R} - \mu_i] = 0$, the first-order term vanishes. □

The lemma shows why the first-order term is governed by deviations of ℓ_i from $1/m$, while the second term is a finite-population ratio-estimator bias.

Proposition 1 (Coherence-controlled RandNLA bound)**.** *Under balanced leverage, for $2 \le k < m$ and $m \ge 4$, there is a constant $K_{c,C} < \infty$, depending only on c and C, such that*

$$\boxed{R_k^{\mathrm{bag},\infty} \le \frac{K_{c,C}}{m\bar{\tau}}\left(V_\delta + V_\delta^2\right) z_k^2.} \tag{2}$$

If V_δ is bounded, this gives

$$R_k^{\mathrm{bag},\infty} = O\!\left(\frac{V_\delta}{m\bar{\tau}} z_k^2\right).$$

For $k \ll m$, $z_k \approx 1/k$, hence

$$R_k^{\mathrm{bag},\infty} = O\!\left(\frac{V_\delta}{m\bar{\tau}k^2}\right).$$

Proof. Balanced leverage implies $L_{i,R} \ge cp$ and $\mu_i \ge cp$. Moreover,

$$\left|\frac{p}{\mu_i} - 1\right| = \frac{|p-\mu_i|}{\mu_i} \le \frac{1}{c}\frac{m-k}{m-1}\frac{|\ell_i - 1/m|}{p} \le K_c|h_i|z_k,$$

for a constant K_c depending only on c.

For the ratio-bias term, the finite-population variance under sampling without replacement gives

$$\mathrm{Var}_R(L_{i,R}) \le \frac{(k-1)(m-k)}{m(m-1)(m-2)}V_\delta.$$

Using $L_{i,R} \ge cp$ and $\mu_i \ge cp$,

$$p\,\mathbb{E}_R\left[\frac{(L_{i,R}-\mu_i)^2}{L_{i,R}\mu_i^2}\right] \le K_c V_\delta z_k.$$

Therefore

$$\left|\frac{\bar{v}_{k,i}}{\ell_i} - 1\right| \le K_c z_k\left(|h_i| + V_\delta\right).$$

Plugging this into (1) and using $\ell_i \le C/m$ yields

$$R_k^{\mathrm{bag},\infty} \le \frac{K_c z_k^2}{A_m}\sum_i \ell_i\left(h_i^2 + V_\delta^2\right) \le \frac{K_{c,C}}{m\bar{\tau}}\left(V_\delta + V_\delta^2\right) z_k^2,$$

which proves the claim. $\square$

Remark 1 (Interpretation)**.** *This is the correct RandNLA message. Uniform subset sampling is close to leverage sampling when leverage scores are balanced. In that regime, the bias of uniform infinite bagging is second order in the sampling distortion, so the excess risk has a z_k^2 envelope. This is the theoretical source of the cube-root rule for the infinite-bagged common-loading case.*

5 Using the low-coherence bound to choose k

Let

$$A = \sigma_\varepsilon^2 + Q^{\mathrm{opt}} = \sigma_\varepsilon^2 + q + \frac{1}{A_m}.$$

The infinite-bagged loss is

$$L_k^{\mathrm{bag},\infty} = (A + R_k^{\mathrm{bag},\infty})\frac{T-1}{T-k}.$$

Under the low-coherence bound, use the envelope

$$R_k^{\mathrm{bag},\infty} \lesssim C_R z_k^2, \qquad C_R \asymp \frac{V_\delta}{m\bar{\tau}}.$$

If $k \ll m$, then $z_k \approx 1/k$ and

$$L_k^{\mathrm{bag},\infty} \approx \left(A + \frac{C_R}{k^2}\right)\frac{T-1}{T-k}.$$

The continuous first-order condition is

$$R_k'(T-k) + (A + R_k) = 0.$$

With $R_k = C_R/k^2$, this becomes

$$Ak^3 + 3C_R k - 2C_R T = 0.$$

For large T,

$$\boxed{k^* \approx \left(\frac{2C_R T}{A}\right)^{1/3}.} \tag{3}$$

Thus the cube-root rule is justified by a low-coherence RandNLA bound. It is not a universal rule.

6 High-coherence case: the safe Jensen bound

If the leverage scores are highly concentrated, the constants in the low-coherence bound explode. This is not a technical nuisance; it is the main economic case where uniform RSM can behave differently. A dominant high-precision forecast means uniform sampling is far from leverage-optimal sampling.

In this regime, do not use a z_k^2 bound unless it is empirically verified. The safe bound is the exact Jensen sandwich.

Proposition 2 (High-coherence fallback bound)**.** *For any positive precision vector* $(\tau_1, \dots, \tau_m)$ *and any* $1 \le k \le m$*,*

$$\boxed{0 \le R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}} \le Q_k^{\text{sub}} - Q^{\text{opt}}.} \tag{4}$$

Under common loading,

$$Q_k^{\text{sub}} - Q^{\text{opt}} = \mathbb{E}_S\left[\frac{1}{A_S}\right] - \frac{1}{A_m}.$$

Thus the high-coherence conservative bound is

$$\boxed{R_k^{\text{bag},\infty} \le \mathbb{E}_S\left[\frac{1}{\sum_{j\in S}\tau_j}\right] - \frac{1}{\sum_{j=1}^m \tau_j}.} \tag{5}$$

Proof. The lower bound follows because w^* minimizes $w'\Sigma_e w$ over all sum-to-one weights, and $\bar{v}_k$ is feasible. The upper bound follows from Jensen's inequality:

$$Q_k^{\text{bag},\infty} = (\mathbb{E}_S v_S)'\Sigma_e(\mathbb{E}_S v_S) \le \mathbb{E}_S[v_S'\Sigma_e v_S] = Q_k^{\text{sub}}.$$

Subtract Q^{opt} from all sides. Under common loading, $Q(S) = q + 1/A_S$ and $Q^{\text{opt}} = q + 1/A_m$, giving (5). □

Remark 2 (Rate implication)**.** *The high-coherence bound is generally of first-order, average-subset type. In many designs,*

$$Q_k^{\text{sub}} - Q^{\text{opt}} \approx C_{\text{sub}}\left(\frac{1}{k} - \frac{1}{m}\right),$$

so the conservative high-coherence envelope is closer to z_k *than to* z_k^2*. This is why the effective exponent can drift from* 2 *toward* 1 *when a few high-precision forecasts dominate.*

7 Choosing k in the high-coherence case

In the high-coherence case, choose k by minimizing a conservative upper bound:

$$L_k^{\text{bag},\infty} \le \left(A + Q_k^{\text{sub}} - Q^{\text{opt}}\right)\frac{T-1}{T-k}.$$

Since under common loading

$$Q_k^{\text{sub}} - Q^{\text{opt}} = \mathbb{E}_S\left[\frac{1}{A_S}\right] - \frac{1}{A_m},$$

one can compute or approximate the conservative rule

$$\boxed{k^*_{\text{cons}} = \arg\min_{1\le k<m}\left[A + \mathbb{E}_S\left(\frac{1}{A_S}\right) - \frac{1}{A_m}\right]\frac{T-1}{T-k}.}$$

See appended validity comments
for suggested wording changes.

If a leading approximation
$$Q_k^{\mathrm{sub}} - Q^{\mathrm{opt}} \approx C_{\mathrm{sub}}/k$$
is used, this returns the usual square-root-type behavior,
$$k^* \asymp \left(\frac{C_{\mathrm{sub}}T}{A}\right)^{1/2}.$$

Thus the practical implication is:
$$\begin{array}{lll} \text{balanced leverage / low coherence} & \Rightarrow & R_k = O(z_k^2), \quad k^* = O(T^{1/3}), \\ \text{high coherence / dominant forecasters} & \Rightarrow & R_k \le Q_k^{\mathrm{sub}} - Q^{\mathrm{opt}}, \quad \text{conservative } k^* = O(T^{1/2}). \end{array}$$

8 Summary

1. Under common loading, the oracle weights are leverage scores:
$$w_i^* = \ell_i = \tau_i/A_m.$$

2. Uniform RSM is a self-normalized uniform subset estimator of the leverage/oracle vector.

3. If leverage is balanced, uniform sampling is close to leverage sampling. Then
$$R_k^{\mathrm{bag},\infty} \le \frac{K_{c,C}}{m\bar{\tau}}(V_\delta + V_\delta^2)z_k^2.$$
For $k \ll m$, this is a $1/k^2$ envelope.

4. If leverage is highly concentrated, do not use the z_k^2 envelope. Use the exact Jensen fallback:
$$0 \le R_k^{\mathrm{bag},\infty} \le Q_k^{\mathrm{sub}} - Q^{\mathrm{opt}} = \mathbb{E}_S[1/A_S] - 1/A_m.$$

5. Therefore the RandNLA argument gives a proper bound only under coherence control. Without such control, the correct high-coherence bound is the average-subset envelope.

Suggested validity edits for the two-sided bounds

These comments are intended to make the last section more precise while preserving the main insight of the note.

1. Proposition 3 (exact two-sided bracket): keep it, but handle the homogeneous case separately.

The exact bracket is strong and worth keeping. A clean statement is:

For non-homogeneous precisions, with $E_k = E_S[1/A_S]$,
$((kE_k - 1/\tau\blacksquare)^2) / (\Sigma_i \tau_i^{-1} - m/\tau\blacksquare) \le R^{bag,\infty}_k \le E_k - 1/A_m$.

Then add a separate sentence: if all τ_i are equal, the denominator is zero and in fact $R^{bag,\infty}_k = 0$ for all k, so the lower bound is interpreted by continuity as 0. This avoids the current 0/0 issue in the homoskedastic case. It is also worth noting that the lower bound is exact at k = 1 and k = m.

2. Corollary 1 (matching z_k forms): label it as a leading-order / asymptotic comparison, not an exact bound.

The exact upper bound is $E_S[1/A_S] - 1/A_m$. The expression $z_k/\tau\blacksquare$ is only the first-order term of a Taylor expansion, because $E_S[1/A_S] = 1/(k\tau\blacksquare) + Var(A_S)/(k\tau\blacksquare)^3 + \cdots$. So the safe wording is: $R^{bag,\infty}_k \le E_S[1/A_S] - 1/A_m = z_k/\tau\blacksquare$ + higher-order positive terms. Similarly, the lower z_k^2 coefficient should be presented as a **large-m leading coefficient**. If you want a finite-m expression, include the extra factor $m/(m-1)^2$ in the lower constant.

3. Corollary 2 (k* bracket): present it as conditional / heuristic unless an extra shape assumption is added.

The exact level bounds on R_k do not by themselves imply a rigorous bound on argmin k*. To turn this into a theorem, add a condition such as: $R^{bag,\infty}_k$ is decreasing and regularly varying in the sense $R_k \blacksquare C z_k^{\alpha}$ for some $\alpha \in [1, 2]$. Under that condition, minimizing $L_k = (A + R_k)(T-1)/(T-k)$ yields $k^* \blacksquare T^{1/(\alpha+1)}$. Hence $\alpha = 2$ gives the cube-root rule, $\alpha = 1$ gives the square-root rule, and the two-sided bracket suggests a range between them. Without this extra regularity condition, the k* bracket should be stated as a motivated heuristic rather than a theorem.

4. Suggested wording for the summary.

A safer final summary is: (i) Proposition 3 gives an exact distribution-free two-sided bracket for $R_k^{bag,\infty}$ under common loading; (ii) the z_k^2 lower side and z_k upper side are leading-order representations of that bracket; (iii) the induced k* range between $T^{1/3}$ and $T^{1/2}$ is conditional on a shape assumption such as $R_k \blacksquare C z_k^{\alpha}$, $\alpha \in [1, 2]$.

In short: Proposition 3 is the strongest part and should stay; Corollary 1 should be marked asymptotic; and Corollary 2 should be stated conditionally unless more shape restrictions are proved.